

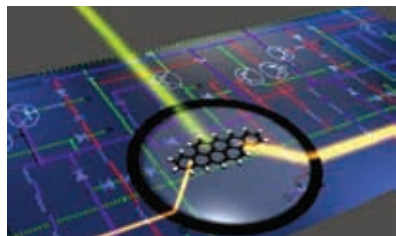
data using quantum mechanics to understand how to perform operations with this data. A quantum nanocomputer would work by storing data in the form of atomic quantum states or spin. Technology of this kind is already under development in the form of single-electron memory (SEM) and quantum dots.

The quantum mechanical properties of atoms or nuclei allow these particles to work together as quantum bits, or qubits. These qubits work together to form the computer's processor and memory and can interact with each other while being isolated from the external environment and this enables them to perform certain calculations much faster than conventional computers.

2. Optical Computers

An optical computer (also called a photonic computer) is a device that performs its computation using photons of visible light or infrared (IR) beams, rather than electrons in an electric current. In most modern computers, electrons travel between transistor switches on metal wires or traces to gather process and store information. The optical computers of the future will instead use photons traveling on optical fibers or thin films to perform these functions.

Currently scientists are focusing on developing hybrids by combining electronics with photonics. This approach appears to offer the best short-term prospects for commercial optical computing, since optical components could be integrated into traditional computers to produce an optical/electronic hybrid NASA. But optoelectronic devices lose about 30% of their energy converting electrons into photons and back and this switching process slows down transmission of messages



Application of Photonics

These hybrid devices will eventually lead to all optical computer systems. All-optical computers eliminate the need for switching. These computers will use multiple frequencies to send information throughout computer as light waves and packets thus not having any electron based systems and needing no conversation from electrical to optical, greatly increasing the speed. 